

Solvent–Molecule Interactions Govern Crystal-Habit Selection in Naphthalene Tetracarboxylic Diimides

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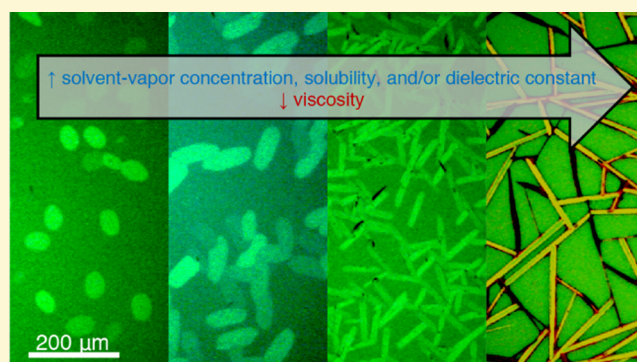
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Supporting Information

ABSTRACT: Controlling hierarchical structural development in organic semiconductors, and across synthetic materials more broadly, is critical to the performance of the material in device applications. Such regulation across multiple scales, from the atomic level to the macroscale, however, is a challenging task given the often-heterogeneous nature of interactions in the processing environment that determines the kinetics and thermodynamics of material growth. Here, we elucidate factors that govern the crystal habit of a core-chlorinated naphthalene diimide (NTCDI-1) and demonstrate the ability to tune its shape in thin films during post-deposition solvent–vapor annealing. Judicious selection of solvent choice and solvent–vapor concentration controls the growth kinetics along different crystallographic axes, and, thus, the resulting habit; we can access isotropic plates that span hundreds of microns to highly anisotropic needles whose long axis can be many millimeters of crystals adopting the same packing polymorph. We find the growth rate along the π -stacking direction of NTCDI-1 during solvent–vapor annealing to scale with its solubility in the solvent and the solvent's viscosity and dielectric constant, with the two former facilitating plasticization. The dielectric constant of the solvent matters because it captures NTCDI-1–solvent interactions. Polar solvents promote π -interactions between neighboring NTCDI-1 molecules, whereas aromatic solvents disrupt these same interactions. Our quantitative understanding of the factors governing crystal-habit selection affords the ability to determine proper post-deposition processing conditions a priori and to access prespecified crystal morphologies in thin films accordingly.



The solid-state packing structures of crystalline molecular materials are intimately connected to their macroscale characteristics that are of interest in prescribed application spaces. Given the intrinsic, hierarchical nature of growth in these materials, changing any aspect of the structure (e.g., molecular packing, crystal size, and shape) can greatly influence material behavior. Differences in crystal habit, for example, can affect properties as wide ranging as catalytic activity,¹ tabletability,^{2,3} dissolution rates,⁴ and solubility.⁵ Yet, the crystal habit is, to first order, governed by the molecular arrangement within the unit cell at quasi-equilibrium conditions. Focusing here on organic semiconductors, molecular materials that adopt two-dimensional (2D) π -stacking arrangements, such as the brickwork^{6–8} or herringbone^{9–11} motifs, tend to form plate-like crystals that facilitate two-dimensional lateral charge transport,¹² whereas those that adopt a one-dimensional (1D) slipped-stack π -stacking motif tend to form needle-like crystals that exhibit highly anisotropic charge transport.^{13–15} While significant efforts in crystal

engineering and structure prediction attempt to elucidate the lower-energy packing configurations and crystal shapes for a given molecular composition and structure, external variables become important in determining the adopted crystal shape, i.e., plates or needles, when crystals are grown far from equilibrium.¹⁶ While there are known methods to access bulk crystals with habits far from equilibrium, e.g., altering experimental growth parameters during solution recrystallization, such as solvent choice,^{17–19} degree of supersaturation,²⁰ or the use of additives,^{21–23} access to such bulk crystals is of limited relevance and generally not suitable for large-area, thin-film electronic applications.

This work aims to elucidate factors and outlines post-deposition processing criteria that control the crystalline domain shape in molecular semiconductor thin films.

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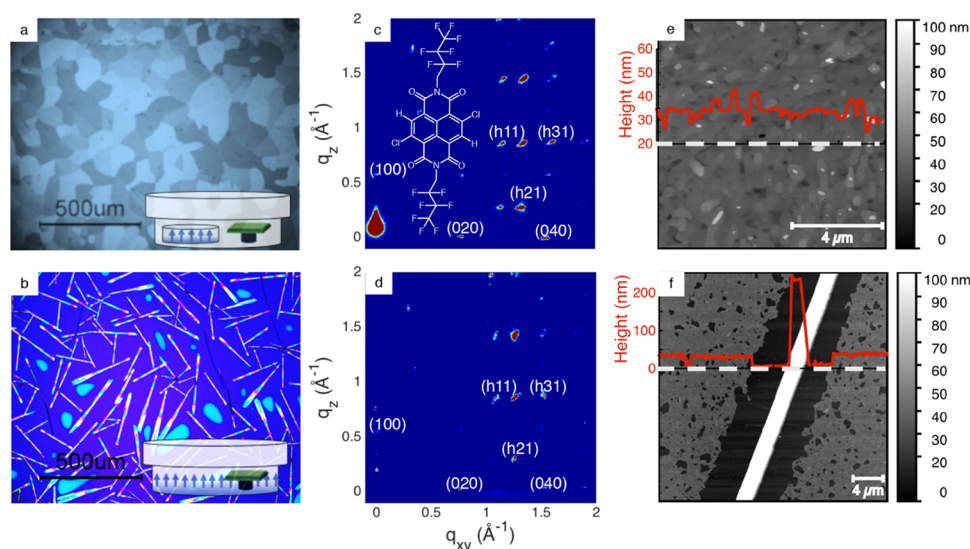


Figure 1. (a) Optical micrograph of an NTCDI-1 film comprised of plate-like domains after acetone–vapor annealing with the setup drawn in the inset. (b) An optical micrograph of an NTCDI-1 film comprised of isolated needle-like crystals after acetone–vapor annealing with the setup drawn in the inset. (c, d) Corresponding 2D GIXD pattern and (e, f) atomic force microscopy (AFM) images obtained from the films imaged in (a) and (b), respectively. Both GIXD patterns are indexed according to the α -phase crystal structure. The insets in (e) and (f) are the height traces of the dashed white line. The needle in (f) is ca. 200 nm tall; the color scale is set to 100 nm for comparison with (e). The chemical structure of NTCDI-1 is shown as an inset in (c).

Molecular semiconductors can be deposited atop large-area substrates via numerous high-throughput techniques. Oftentimes, the constraints of rapid deposition kinetically trap such films in undesired morphologies far from equilibrium.²⁴ Subjecting these films to post-deposition processing can unlock the arrested structure and further direct solid-state structural development.^{25,26} 2,7-Dioctyl[1]benzothieno[3,2*b*]-[1]benzothiophene (C8 BTBT), for instance, forms small, plate-like grains on spin-coating; exposing these films to solvent vapor results in the growth of isolated needle-like single crystals. Transistors comprising these needle-like crystals exhibit mobilities that are an order of magnitude higher than those of transistors comprising the as-deposited polycrystalline films.²⁷ As another example, rubrene (5,6,11,12-tetraphenyltetracene) forms amorphous films when deposited via thermal evaporation and crystallizes upon post-deposition thermal annealing, with the choice of the annealing temperature governing the preferential formation of grains over spherulites.^{28,29}

Here, we explore the use of solvent–vapor annealing to control the crystal habit in thin films of a core-chlorinated naphthalene diimide derivative, NTCDI-1, during its β -to- α polymorphic transformation. We previously demonstrated that this transformation follows a nucleation and growth mechanism with α -phase plates growing at the expense of β -phase grains during solvent–vapor annealing.³⁰ With α -phase NTCDI-1 exhibiting a 1D slipped-stack packing motif and crystals adopting a needle habit during solution recrystallization,¹³ we expected the polymorphic transformation to result in needle-like domains, rather than the plate-like domains we observed.³⁰ Interestingly, we found that the adopted crystal habit and accompanying β -to- α transformation kinetics during solvent–vapor annealing depend on the solvent–vapor concentration and solvent type, respectively. Here, we quantitatively elucidate the mechanism by which crystal-habit selection takes place during this β -to- α polymorphic transformation of NTCDI-1.

Figure 1 shows the optical micrographs of NTCDI-1 films after solvent–vapor annealing at different acetone–vapor concentrations. Plates with an in-plane length-to-width aspect ratio of near unity are formed throughout the film (Figure 1a) when annealing is conducted by placing a small acetone reservoir inside a larger chamber (see the inset). Directly placing acetone in the larger chamber increases the surface area of the solvent by a factor of three; exposing NTCDI-1 thin films to this increased solvent–vapor concentration results instead in the formation of isolated needle-like crystals (Figure 1b). Despite the drastically different macroscopic crystal habits, grazing-incidence X-ray diffraction (GIXD) measurements indicate that both plates and needles of NTCDI-1 comprise the α -phase polymorph, with the a -axis of its unit cell preferentially oriented normal to the substrate. Because the plates and needles exhibit the same preferential out-of-plane orientation, differences in the crystal habit must stem from differences in growth rates in-plane, i.e., along the b - and c -axes of the α -phase. The Bravais, Friedel, Donnay, and Harker (BFDH)^{31–33} crystal morphology of this plane is shown in Figure S1; we see that intermolecular π – π interactions dominate the molecular packing of this b – c crystallographic plane, with strong π -stacking occurring along the c -axis and little-to-no π -stacking along the b -axis. Given the anisotropy in the strength of molecular interactions along these two crystallographic directions, we expect growth to be the fastest along the c -axis. This expectation is consistent with accessing needle-like crystals when the α -phase NTCDI-1 is grown under quasi-equilibrium conditions. We thus surmise that our access of plate-like domains during NTCDI-1's solid–solid phase transformation on solvent–vapor annealing to stem from factors limiting growth along the π -stacking direction.

Figure 1e,f shows the atomic force microscopy (AFM) images of NTCDI-1 plates and needles, respectively. The surface of the plate-like domain has an RMS roughness of 3.5 nm; the step changes are consistent with terraces having different numbers of molecular layers, with each molecular

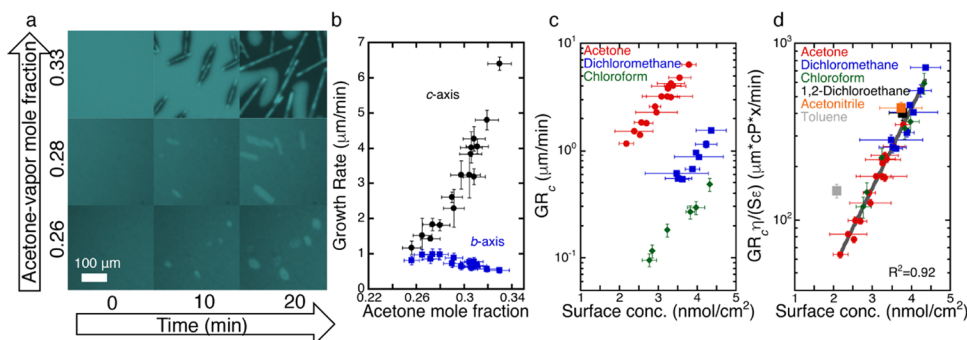


Figure 2. (a) Time-resolved optical micrographs during in situ acetone–vapor annealing at different acetone–vapor concentrations. (b) Growth rate along the different in-plane crystallographic axes as a function of acetone–vapor concentration. (c) Growth rate along the *c*-axis (GR_c) of the α -phase NTCDI-1 domains during solvent–vapor annealing varies dramatically with both solvent–vapor concentration and solvent choice. (d) Scaling the growth rate by solubility ($S [=]$ mole NTCDI-1 per mole solvent, denoted as x), the solvent viscosity (η), and dielectric constant (ϵ) of the solvent collapses the data for all solvents tested in this study.

layer approximately 1.2 nm in height. Given that the thickness of the films is invariant at 80 nm with solvent–vapor annealing (see Figure S2), we surmise that the β -to- α transformation in this case primarily involves local rearrangement of NTCDI-1 without any substantive molecular diffusion. The needle-like domain in Figure 1f, on the other hand, is approximately 250 nm tall, or three times taller than the initial height of the film. We deduce that substantial molecular diffusion occurs when the β -to- α transformation leads to needles; vertical growth along the *a*-axis occurs by depleting NTCDI-1 that immediately surrounds the needles, resulting in gradual isolation of these anisotropic domains on the surface of the silicon substrate. While these needles can be several times taller than the height of the original films, this height increase in the out-of-plane direction is still small compared to the needle length, which can often span many millimeters. As such, our subsequent analysis only considered how differences in the in-plane growth lead to plates versus needles.

To systematically investigate the influence of solvent–vapor concentration on crystal habit during annealing, we constructed a flow setup (see Figure S3) equipped with mass-flow controllers with which we can tune the solvent–vapor concentration through dilution with nitrogen. Figure 2a shows optical micrographs collected at different time points during solvent–vapor annealing at three different acetone–vapor concentrations. As the acetone mole fraction increases from 0.26 to 0.33 (the balance is N_2), the crystal habit becomes more anisotropic, evolving from plate-like to needle-like. In all cases, the crystalline domains uniformly exhibit a single coloration; the absence of birefringence under the polarized optical microscope suggests that these domains individually adopt a single in-plane molecular orientation, enabling quantification of growth rates within the *b*–*c* crystallographic plane. Figure 2b depicts the growth rate along both the *b*- and *c*-axes as a function of acetone–vapor mole fraction. As the acetone mole fraction increases from 0.26 to 0.33, the growth rate along *c*-axis increases from 1 to almost 7 $\mu\text{m}/\text{min}$, while the growth rate along the *b*-axis remains largely invariant; we attribute the slight decrease in the growth rate along the *b*-axis at high solvent–vapor concentrations to the depletion of NTCDI-1 surrounding individual needles. It follows that differences in the crystal habit must stem from variations in the growth rate along the *c*-axis of α -phase NTCDI-1 with solvent–vapor concentration, as both the absolute growth rates and the differences in growth rates along

the *a*- and *b*-axes with solvent–vapor concentration are small compared with those along the *c*-axis. The fact that differences in the growth rate along the *c*-axis dictate the crystal habit adopted by NTCDI-1 α -phase is further evidenced when we alter the acetone–vapor concentration in situ during a single annealing experiment. The series of optical micrographs in Figure S4 shows how we can tune the growth mode from 1D to 2D, and back to 1D as we decrease, and then increase, the solvent–vapor concentration. Collectively, these time-resolved experiments suggest that increased solvent–vapor concentration leads to increased plasticization, providing NTCDI-1 with ample mobility to access its thermodynamically favored needle-like crystal habit.

In addition to solvent–vapor concentration, we assessed the impact of solvent choice on the kinetics and the resulting crystal habit of this polymorphic transformation. We chose to investigate dichloromethane (DCM) and chloroform due to their high vapor pressures. Figure S5 shows the optical micrographs of films annealed with different vapor concentrations of DCM or chloroform for 1.5 and 4 h, respectively. Reducing the vapor concentrations of both DCM and chloroform results in NTCDI-1 adopting α -phase domains that are more plate-like, qualitatively similar to what we observed when NTCDI-1 was annealed with low concentrations of acetone vapor. To properly compare results from different solvents, we employed quartz crystal microgravimetry (QCM) to estimate the concentration of solvent either adsorbed on the surface of, or penetrated into, the film via the Sauerbrey equation (see Figure S6).³⁴ We refer to this concentration as the surface concentration, recognizing that the solvent may have penetrated the film. Figure 2c tracks the growth rate along the *c*-axis of α -phase NTCDI-1 domains as a function of the QCM-determined surface concentration when acetone, DCM, and chloroform are employed during annealing. Irrespective of the solvent choice, the growth rates along the *c*-axis of the α -phase NTCDI-1 domains exhibit a strong dependence on the solvent–vapor concentration, with needles accessed at high solvent–vapor concentrations. This observation is consistent with our picture that the quasi-equilibrium needle-like crystal habit is only accessed when NTCDI-1 has sufficient mobility. Solvent–vapor annealing at low concentrations limits solvent penetration of the film and, consequently, NTCDI-1 molecular diffusion, resulting in a retardation in growth along the *c*-axis and access to more isotropic plate-like domains. Importantly, the growth rates vary

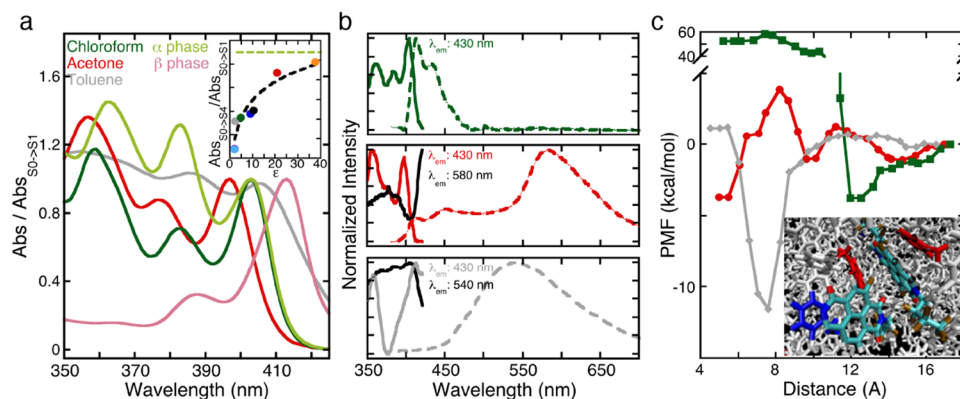


Figure 3. (a) Absorption spectra for 1 μ M NTCDI-1 solutions in representative solvents and thin films of α - and β -phase NTCDI-1. All spectra have been normalized to the intensity of the peak of the low-energy transition, identified in the TDDFT calculations as the $S_0 \rightarrow S_1$ transition (ca. 400 nm). The intensity (relative to the lowest-energy transition) of the $S_0 \rightarrow S_3$ transition, shown in the inset, increases monotonically as a function of solvent dielectric constant, approaching the value of the α -phase film. (b) Excitation and emission ($\lambda_{\text{exc}} = 360$ nm) spectra for NTCDI-1 in chloroform (green), acetone (red), and toluene (gray). Excitation scans tracking aggregate emission are shown as black traces. (c) The potential of mean force (PMF, kcal/mol) between two NTCDI-1 molecules is shown as a function of center-to-center distance. The inset shows a snapshot from steered molecular dynamics (SMD) simulations of NTCDI-1 (aqua) in a sea of toluene (gray) at its free energy minimum. The toluene molecules closest to the NTCDI-1 molecules are colored in red and blue. The red toluene molecules sandwich one NTCDI-1 molecule; the co-facial arrangement between toluene and NTCDI-1 within the sandwich structure is highlighted by the blue toluene molecule.

quantitatively at similar solvent–vapor concentrations for the different solvents tested, indicating specificity of NTCDI-1’s polymorphic transformation with solvent choice.

To understand the underlying solvent–molecule interactions, we searched for scaling parameters to collapse the growth rates across different solvent choice universally. While scaling by any single obvious solvent parameter that would impact solvent–molecule interactions did not collapse our data, we found scaling the growth rate by the solubility of NTCDI-1 in the particular solvent, the solvent’s dielectric constant, and viscosity to collapse the data across solvent choice (Figure 2d). To assess the robustness of this scaling, we conducted additional solvent–vapor annealing experiments with toluene ($\epsilon = 2.38$; $\eta = 0.59$ cP), 1,2-dichloroethane (DCE; $\epsilon = 10.36$; $\eta = 0.79$ cP), and acetonitrile ($\epsilon = 37.5$; $\eta = 0.37$ cP), given that these solvents span a wide range of dielectric constants and viscosities. Figure S7 shows the optical micrographs of NTCDI-1 films after solvent–vapor annealing with toluene, DCE, and acetonitrile. The absolute growth rates of NTCDI-1 during solvent–vapor annealing vary dramatically across the different solvents. With the exception of the growth rate during toluene–vapor annealing, the growth rates during annealing with other solvent vapors collapse uniformly with the scaling argument above (see Figure 2d). We observe that the scaled growth rate with toluene–vapor annealing deviates from all other scaled growth rates by a factor of approximately two, which we attribute to increased π – π interactions between NTCDI-1 and toluene given its aromatic nature, which is discussed below. Given this analysis, we believe this scaling to be generally applicable across a wide range of non- and weakly interacting solvents, giving rise to a useful design principle governing the solvent–vapor annealing conditions by which we can dial in prescribed growth rates of α -phase NTCDI-1 domains.

The strong correlation between the NTCDI-1 growth rates and these different solvent parameters should afford insight into the solvent–molecule interactions governing this polymorphic transformation. To properly unravel the origin behind the scaled parameter, we would need to be able to change one

variable at a time at a constant solvent–vapor concentration across multiple solvents, which we found to be nontrivial. As such, we do not speculate at the physical origin of the scaled parameter but rather rationalize how the individual components could influence crystal growth during solvent–vapor annealing. The solubility of NTCDI-1 ranges from 2.9×10^{-4} moles NTCDI-1/mole in acetone and toluene, to 6.0×10^{-5} moles NTCDI-1/mole in acetonitrile (see Table S1). Given that “like dissolves like”, any increase in solubility should result in enhanced plasticization, and, thus, transformation, of NTCDI-1 films. Supporting this hypothesis is that antisolvents of NTCDI-1 do not induce any structural change upon solvent–vapor annealing. Like solubility, viscosity is involved with film plasticization. We think of viscosity as a measure of internal friction between molecules; solvents with higher viscosity should slow plasticization, slowing the transformation kinetics. That the growth rate scales with the dielectric constant of the solvent implicates the importance of solvent–molecule interactions during polymorphic transformation. The charge distribution of NTCDI-1 (shown in Figure S8) highlights a largely neutral aromatic face and oxygens on the periphery of the molecule that are negatively charged, giving rise to the quadrupolar nature of NTCDI. It follows that polar solvents should preferentially interact with the charged oxygen atoms and fluoroalkyl chains on NTCDI-1, whereas nonpolar solvents should preferentially interact with the largely neutral aromatic core. We thus surmise that employing polar solvents during solvent–vapor annealing of NTCDI-1 should enhance π – π interactions between neighboring NTCDI-1 compared to when nonpolar solvents are used, whose interactions with the aromatic face of NTCDI-1 might hinder π – π interactions between neighboring NTCDI-1. As such, we expect the growth rate along the c -axis of NTCDI-1 to be fastest when annealing is conducted with solvents with high dielectric constants.

To further understand the nature of these solvent–molecule interactions, we turned to a combination of optical spectroscopy and theoretical calculations of NTCDI-1 in these different solvents, recognizing that we are inferring the nature of molecular interactions in the solid state by extrapolating

from their characteristics in solution. Figure 3a shows the UV–vis absorption spectra of 1 μ M solutions of NTCDI-1 in acetone, chloroform, and toluene, along with the solid-state absorption spectra of the α - and β -phases of NTCDI-1. Acetone, chloroform, and toluene were chosen as representative examples for polar, nonpolar, and aromatic nonpolar solvents, respectively; the absorption spectra of NTCDI-1 in additional solvents of varying polarity are shown in Figure S9. In general, the absorption spectrum can be considered a fingerprint of a chromophore's local environment, i.e., if a chromophore excitonically couples with its molecular neighbors. Variations in solvent polarity, which will alter these short-range interactions, will thus manifest changes in the absorption spectrum when compared to that of the isolated chromophore. Qualitatively, the absorption spectra of the NTCDI-1 α - and β -phases are quite different, with the latter showcasing vibronic characteristics expected for an isolated pyrene-based molecule that are absent in the former. The absence of the vibronic features in the α -phase spectrum is consistent with NTCDI-1 adopting more prominent co-facial core-to-core (π – π) stacking than in the β -phase, resulting in stronger excitonic coupling between neighboring chromophores. Interestingly, the solution spectra of NTCDI-1 tend to resemble that of the α -phase more than that of the β -phase independent of solvent choice.

The inset shown in Figure 3a tracks the relative intensity of two NTCDI-1 optical transitions, at ca. 400 and 360 nm, as a function of the solvent dielectric constant; these optical transitions are identified as the $S_0 \rightarrow S_1$ transition and $S_0 \rightarrow S_4$ transition, respectively, through density functional theory (DFT) and time-dependent DFT (TDDFT) calculations (see Table S2). Our TDDFT calculations indicate that this ratio should remain nearly constant with dielectric constant, yet we experimentally observe this ratio to increase with increasing solvent dielectric constant, approaching the value corresponding to that extracted from the absorbance spectrum of the α -phase film. These results suggest that the variations observed in the experimental spectra are not due to dielectric effects alone but are likely due to differences in intermolecular interactions that give rise to varying extents of aggregation of NTCDI-1. Polar solvents are likely to interact with both the peripheral ketones and fluoroalkyl chains of NTCDI-1, increasing the tendency for NTCDI-1 to self-aggregate through core π – π interactions, while nonpolar solvents are likely to remain dispersed around the NTCDI-1 core, keeping NTCDI-1 well-solvated. We surmise that the solution spectra exhibit both isolated and aggregated NTCDI-1 character, with an increasing oscillator strength of the higher-energy transitions with increasing solvent polarity to indicate stronger NTCDI-1 aggregation in solution.

The different coloration of NTCDI-1 solutions (Figure S10) is consistent with this picture of NTCDI-1 adopting varying extents of aggregation in different solvents. Solutions of NTCDI-1 in acetone or acetonitrile exhibit a faint pink hue, whereas solutions of NTCDI-1 in nonpolar chlorinated solvents (DCM, CF, 1,2-DCE) are colorless and solutions of NTCDI-1 in toluene are bright yellow in color. Figure 3c shows the excitation and emission spectra of NTCDI-1 in acetone, chloroform, and toluene (see Figure S9 for emission spectra of NTCDI-1 in other solvents). We observe features associated with molecular emission of NTCDI-1 at ca. 410 and 430 nm, and broader emission from aggregated NTCDI-1 from 500 to 700 nm, the extent of which varies in each solvent.

Excitation scans track both molecular and aggregate emission; with the exception of toluene, the excitation scans tracking the molecular emission at 430 nm resemble the absorption spectra in their peak positions, shapes, and amplitudes for NTCDI-1 in each of the solvents. In toluene, however, it is instead the excitation scan of NTCDI-1 tracking the aggregate emission that more closely resembles its absorption spectrum, suggesting that toluene and NTCDI-1 form an excitonically coupled solvent–molecule aggregate.^{35,36} Strengthening our hypothesis is the observation that the same emission characteristics and yellow coloration of the solution are observed when NTCDI-1 is dissolved in *p*-xylene, with this aggregate emission distinct from that observed in the spectra of NTCDI-1 in acetonitrile and acetone.

To better understand the nature of the NTCDI-1 aggregates in solution, we conducted steered molecular dynamics (SMD) free energy calculations for two NTCDI-1 molecules in a sea of solvent molecules. Thermodynamic integration (TI) calculations provide the free energy along the path, shown as the potential of mean force (PMF), for NTCDI-1 in chloroform, acetone, and toluene in Figure 3c. The PMF of NTCDI-1 in chloroform experiences a large drop in free energy when NTCDI-1 is separated by more than 12 Å, consistent with the picture of NTCDI-1 being well-solvated in chloroform. The PMF is qualitatively different for NTCDI-1 in acetone compared to chloroform. The PMF exhibits minima at the initial configuration and when the molecules are separated by a large distance, suggesting the presence of both self-aggregated and solvated NTCDI-1 in acetone. In toluene, the PMF of NTCDI-1 exhibits a minimum in the free energy only when the NTCDI-1 molecules have a core-to-core distance of ca. 7 Å. The inset in Figure 3c shows a snapshot of the SMD simulation at this energy minimum, revealing toluene and NTCDI-1 present in a sandwich-type aggregate that precludes π – π interactions between NTCDI-1 molecules. Collectively, a picture of how solvent polarity tunes the relative strength of molecule–molecule and molecule–solvent interactions emerges: NTCDI-1 is well-solvated in nonpolar solvents (chloroform), self-aggregates in polar solvents (e.g., acetone), and readily forms solvent–molecule aggregates with aromatic solvents through hetero- π – π interactions (toluene). In extending these findings to solvent–vapor annealing of NTCDI-1, we believe that polar solvents promote π – π stacking of NTCDI-1 molecules; the growth rate along the *c*-axis, or π -stacking direction, of the α -phase is, thus, the highest when polar solvents are employed during solvent–vapor annealing. Because aromatic solvents interact with the NTCDI-1 aromatic core, we observe retarded growth along its π -stacking direction. Our experiments suggest that understanding the molecular interactions at play in these systems at a microscopic level can provide substantial insight into their macroscopic solid-state structural development.

We demonstrate the ability to control the crystal habit of α -phase NTCDI-1 thin films via post-deposition solvent–vapor annealing through tuning the solvent–vapor concentration and solvent choice. By understanding the solvent–molecule interactions at play and the mechanism by which solvent–vapor annealing takes place, we can increase our predictive power for choosing appropriate processing conditions to access prespecified morphologies. Irrespective of solvent, increasing solvent–vapor concentration increases the extent of plasticization. This enhanced molecular mobility promotes growth along the fast growth axis of α -phase NTCDI-1 and results in

the formation of needle-like domains. The growth rate can further be increased by using solvents with low viscosities and high dielectric constants. The former increases the solvent's mass transfer through the film via the Stokes–Einstein relation, while the latter promotes specific solvent–molecule interactions that promote the NTCDI-1 molecular arrangement present in the α -phase polymorph. Employing low-viscosity polar solvents at high solvent–vapor concentrations can, thus, result in the formation of isolated single-crystalline needles. While not the focus of this manuscript, controlling nucleation by patterning substrates prior to deposition should result in large arrays of single crystals that could be individually addressable in electronic circuits.^{37,38} Our experiments demonstrate how kinetic limitations can drastically alter the crystal habit away from their equilibrium forms; understanding the governing parameters allows precise control over the solid-state structure.

EXPERIMENTAL METHODS

Materials. All solvents were purchased from Fisher Scientific and used as-received. The synthesis of NTCDI-1 followed reported procedures in the literature without further modification.¹³

Thin-Film Deposition. NTCDI-1 was thermally evaporated at a rate of $0.5\text{--}1\text{ Å s}^{-1}$ under a base pressure of 1×10^{-6} mbar to form a uniform film with a thickness of 80 nm atop Si(100) wafers with a 300 nm thermally grown SiO_2 layer (Process Specialties, Inc.). The substrates were sequentially sonicated in deionized water, acetone, and isopropanol and dried with nitrogen prior to NTCDI-1 deposition.

Structural Characterization. Optical micrographs were collected using a Nikon Eclipse ME600 microscope equipped with a Nikon DXM 1200 camera. GIXD experiments were conducted at the G1 (9.95 ± 0.05 keV) station at the Cornell High Energy Synchrotron Source. The scattered intensity was collected using a two-dimensional CCD detector (Finger Lakes Instrumentation). All GIXD images have been background subtracted. AFM images were collected using a Bruker NanoMan AFM with the Dimension platform and NanoScope V controller. All AFM images were processed using Gwyddion (v2.5). Fluorescence spectroscopy was performed on a Horiba PTI QuantaMaster 400 spectrofluorometer. Absorption measurements were collected with an Agilent Technologies Cary 5000 spectrophotometer.

Growth Rate Determination. Time-series of optical images (typically $10\times$ magnification) were collected during in situ solvent–vapor annealing experiments. The growth rate of individual crystals was determined by tracking the crystal size (measured in pixels) as a function of time, which was found to have a linear correlation. The growth rate was determined for crystals prior to impingement on other crystals (i.e., no boundary effects).

Density Functional Theory (DFT) Calculations. All density functional theory (DFT) calculations were carried out using the Gaussian09 (Rev. E.01) software suite.³⁹ To examine the nature of the optically excited states in a variety of solvents, optimizations of the neutral, ground-state geometry were undertaken at the PBE0/6-311g(d,p) level of theory; we note that the alkyl chains were truncated after the perfluoromethyl group near the NTCDI core to reduce the computational cost. The polarizable continuum model (PCM) within the self-consistent reaction field (SCRF) theory was used to implicitly account for the dielectric of the of the acetonitrile ($\epsilon = 35.688$), acetone ($\epsilon = 20.493$), 1,2-dichloroethane ($\epsilon = 10.125$), dichloromethane ($\epsilon = 8.93$), chloroform ($\epsilon = 4.7113$), and toluene ($\epsilon = 2.3741$) solvent environments in which the UV–vis experiments were carried out. All geometries were confirmed as minima of the respective potential energy surfaces through normal mode analyses. Time-dependent DFT (TDDFT) calculations were then performed, again within PCM, to determine the vertical excitation energies. Electronic couplings among dimers directly extracted from the NTCDI-1 α crystal (i.e., no further geometry relaxations were carried

out) were determined through the fragment orbital approach at the $\omega\text{B97XD}/6\text{-}31\text{G(d,p)}$ level of theory.⁴⁰

Point-Charge Distribution. We used ORCA to calculate CHELPG charges for our models in the molecular dynamics (MD) simulation because they are known to work well with the OPLS force field.⁴¹

Free-Energy Calculations. We used a steered molecular dynamics (SMD) approach⁴² as implemented in LAMMPS to calculate the free energy of different configurational states of two NTCDI-1 molecules in different solvents. The initial configuration of the two NTCDI molecules was geometry optimized in ORCA. We coupled their centers of mass with a spring and pulled them apart with a relative speed of 1 Å/ns . The spring constant of the spring was 100 kcal/(mol Å) , and the direction of the spring was set to “auto” within LAMMPS so that it was able to explore more fully a thermodynamically favorable pathway. We used an isobaric–isothermal (NPT) ensemble with the temperature set at 298.15 K and pressure at 1 atm . We ran each NPT simulation for 20 ns and output the forces executed on each NTCDI molecule. We used the exact same settings across all three solvent systems. Moreover, to accurately sample the forces for thermodynamic integration, we separated the trajectories of NTCDI molecules into 24 windows and ran individual simulations on each window. We were mainly interested in the behavior of the $\pi\text{--}\pi$ stacking of the aromatic carbon atoms (we refer to these atoms as the NTCDI “core”). In each simulation, we fixed the position of the NTCDI cores and output the forces required to maintain such a conformation. We ran each simulation at least a further 5 ns to allow the solvent and the side-chains of NTCDI to fully relax. The difference between the forces on NTCDI molecules 1 and 2 was the ensemble force average at each reaction coordinate. The distance between NTCDI molecules was calculated based on the center of mass of the NTCDI cores. These calculations are defined as follows

$$f(\lambda_n) = f(\lambda_n)\text{NTCDI}_1 - f(\lambda_n)\text{NTCDI}_2$$

where λ_n is the reaction coordinate and defined as the difference between the centers of mass of two NTCDI molecules and n represents each window of thermodynamic integration.

We used the trapezoid rule to integrate the force along the reaction pathway to calculate the free energy defined as follows

$$F_n = \int_{\lambda_0}^{\lambda_n} \langle f(\lambda) \rangle_\lambda d\lambda \approx \sum_{i=1}^n \frac{1}{2} (f(\lambda_{i-1}) + f(\lambda_i)) \Delta\lambda_i$$

where $i = 1, 2, \dots, n$ and $\langle \dots \rangle$ denotes an ensemble average.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.chemmater.9b03142>.

Additional experimental and computational results highlighted in the main text, including micrographs characterizing thin-film morphology and results from TDDFT calculations (PDF)

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The manuscript was written through contributions of all authors.

Notes

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